Spatial data input and manipulation

Lecture 3 content:

- Different spatial data capture techniques.
- The fundamentals of a GNSS.
- Spatial data formats and transfer.
- Basics of spatial data editing.
- Assessing spatial data quality.
- Some of the main errors in spatial data.

Spatial data capture

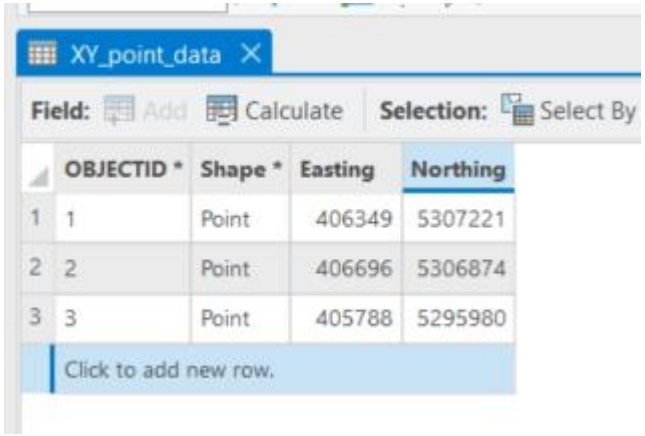
- Spatial data capture is the process of generating new digital spatial data.
- It involves direct measurement or analogue-to-digital conversion.

Vector data capture

- keyboard entry
- digitising
- field surveying
- photogrammetry
- LiDAR (light detection and ranging)

Keyboard entry

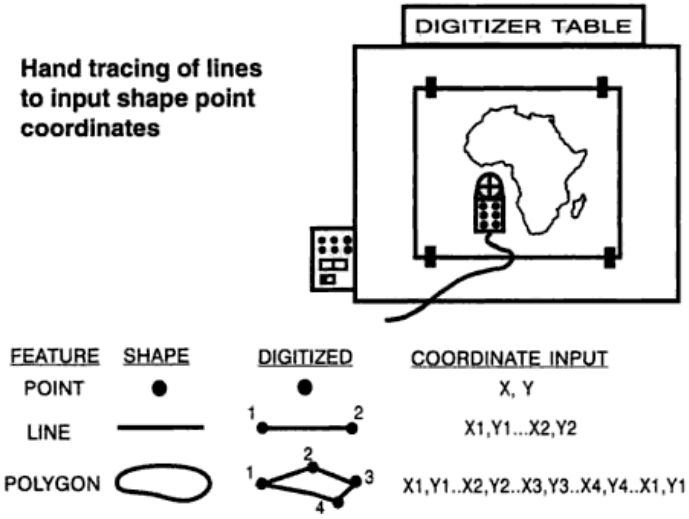
- Involves obtaining the coordinates of point features from maps, field data or other source documents, and typing them into the GIS.
- Point features are also called *XY events.*
- Used when the number of point features is small.



Manual digitising

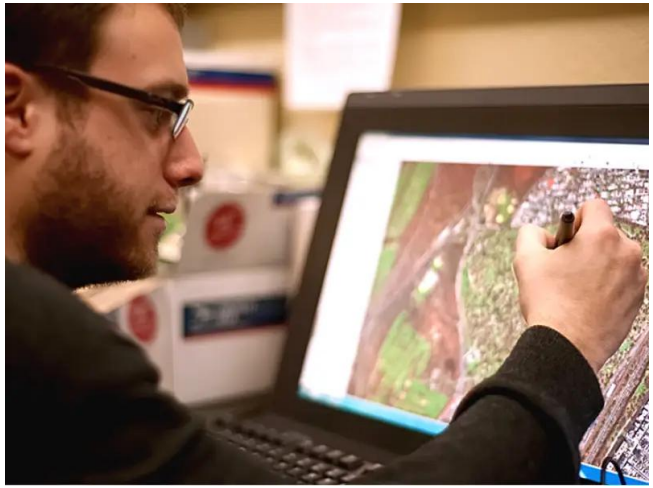
- Manually tracing along boundaries of geographical features from a map or image, while recording the coordinates of the points selected to approximate the boundary or shape of each feature.
- Digitizing table and a puck.

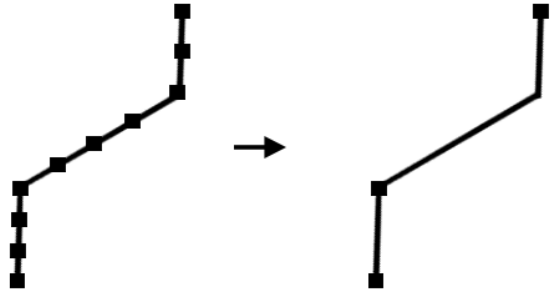




On-Screen digitising

- On-screen digitising using scanned paper maps as digital images and then tracing the features on a computer screen.
- Limited by the resolution of the scanned images and may not capture fine details as accurately as manual digitizing.

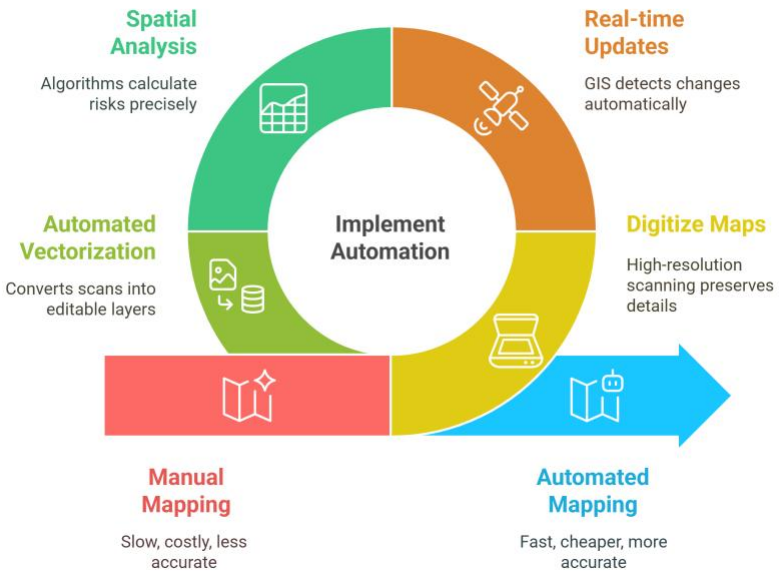




Digitising involves accuracy not density of points

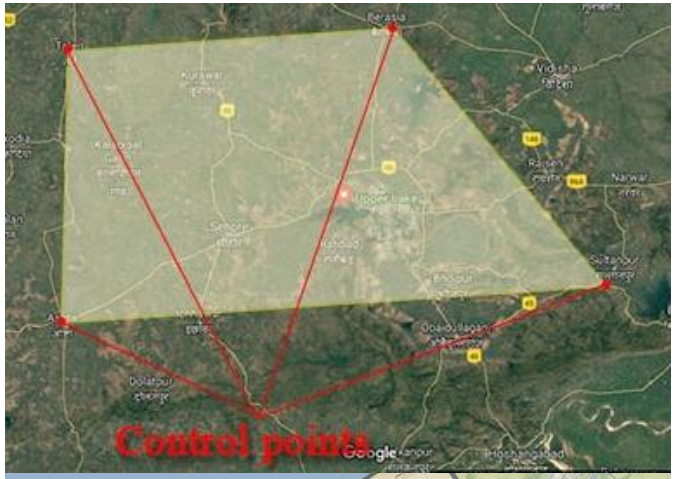
Automatic digitising

- Automatic digitizing converts raster images into vector data using raster-to-vector conversion.
- This method is highly efficient for handling large datasets.
- Part of a wider push toward automated digital mapping.



Digitising data in GIS







https://aigeo360.com/digitization-in-gis/

Field surveying

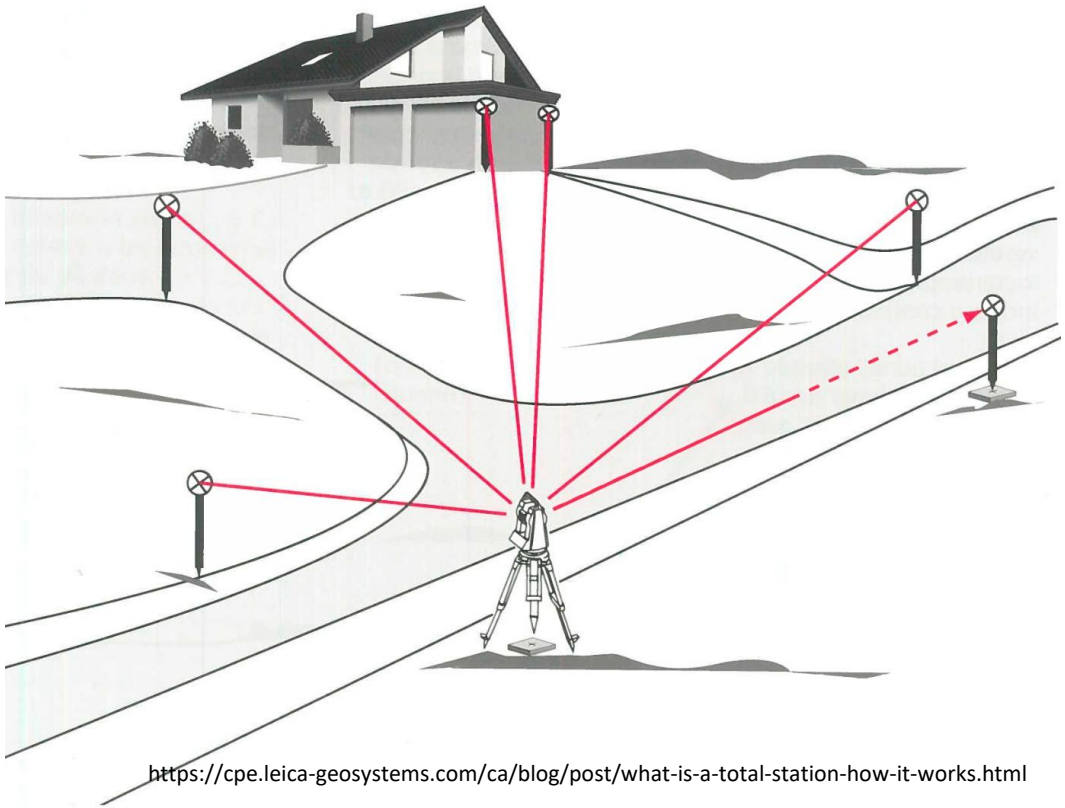
- Collect spatial data through direct measurements in the field.
- Traditionally surveying methods are used
- More recently, total stations and GNSS.







Total station surveying



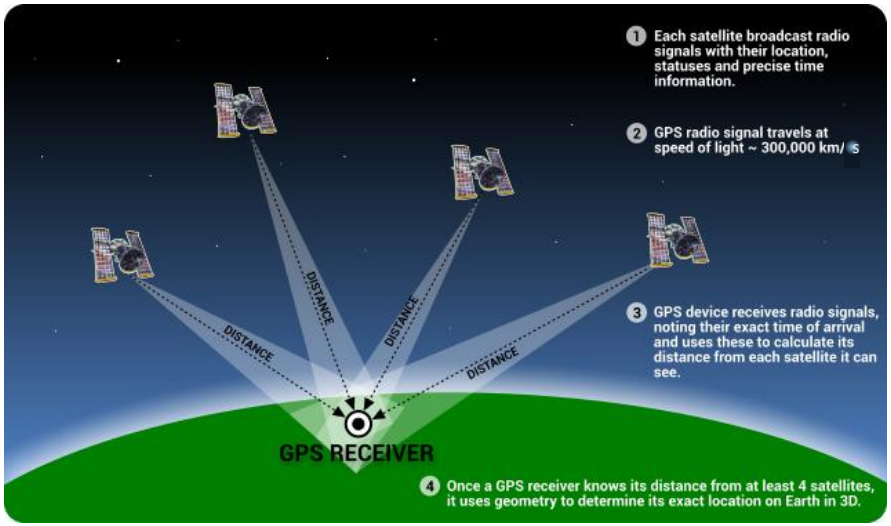
https://cpe.leica-geosystems.com/ca/blog/post/what-is-a-total-station-how-it-works.html



https://www.clevelandbrothers.com/job-site-layout-base-station-total-station/

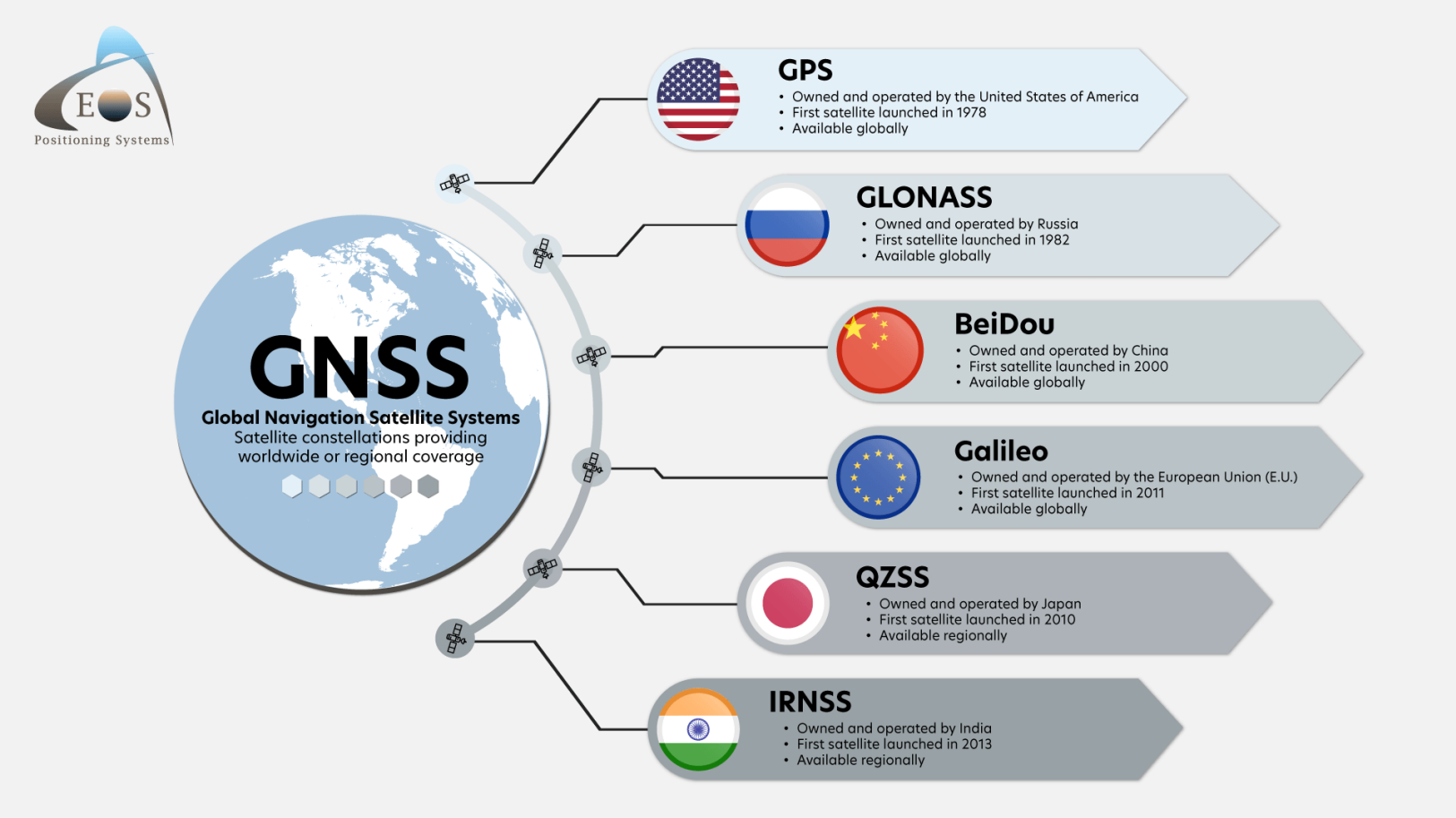
Global Navigation Satellite System (GNSS)

- Provides accurate measurement of a position almost anywhere on the surface of the Earth.
- Hand-held GNSS receivers allow data collectors to walk or drive around to collect (*x*, *y*, *z*) coordinates at sample points.
- GPS (USA), GLONASS Russia), Galileo (Europe) and BeiDou (China) are three main GNSS systems that are currently operational.



https://vangarde-industries.com/products/gps-tracking-system

GNSS constellations with global coverage



https://eos-gnss.com/knowledge-base/gps-overview-1-what-is-gps-and-gnss-positioning

Main features of a GNSS

- GPS satellites orbit 20,200km above the Earth in a period of twelve hours in six orbital planes.
- This configuration ensures that at least **four** satellites (the minimum number required for accurate measurements) are always potentially visible anywhere on Earth
- Each satellite emits signals in 2-3 frequency bands as radio waves, long wavelength electromagnetic radiation which is able to penetrate the atmosphere. At least one of the bands is reserved for military use.
- The low power means arriving signals are not much different from background noise level.

Smartphone GNSS

Alt = altitude above WGS84 ellipsoid.

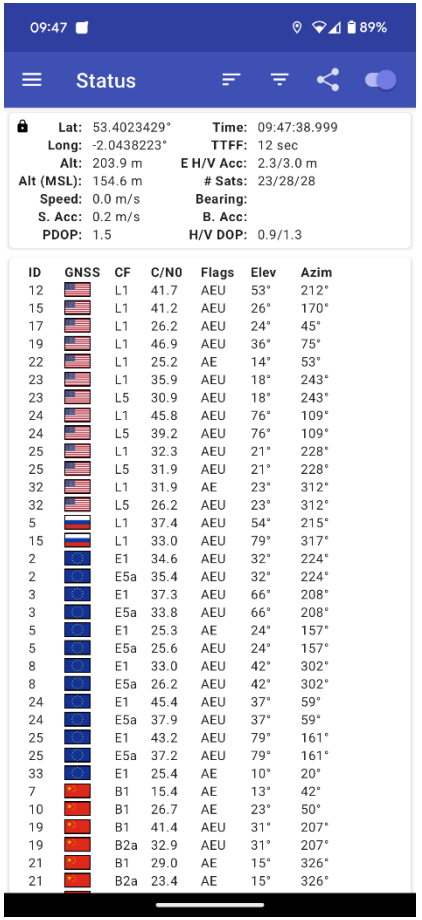
Alt (MSL) = geoid-based Earth Gravitational Model (e.g. EGM96 which defines sea level).

TTFF = Time To First Fix on satellites.

E H/V Acc = Estimated Horizontal and Vertical accuracy.

L1/L5 radio bands (L5 civilian high precision)

GPStest app for Android downloaded from Google Play



ID	GNSS	CF	C/N0	Flags	Elev	Azim
12		L1	41.7	AEU	53°	212°
15		L1	41.2	AEU	26°	170°
17		L1	26.2	AEU	24°	45°
19		L1	46.9	AEU	36°	75°
22		L1	25.2	AE	14°	53°
23		L1	35.9	AEU	18°	243°
23		L5	30.9	AEU	18°	243°
24		L1	45.8	AEU	76°	109°
24		L5	39.2	AEU	76°	109°
25		L1	32.3	AEU	21°	228°
25		L5	31.9	AEU	21°	228°
32		L1	31.9	AE	23°	312°
32		L5	26.2	AEU	23°	312°
5		L1	37.4	AEU	54°	215°
15		L1	33.0	AEU	79°	317°
2		E1	34.6	AEU	32°	224°
2		E5a	35.4	AEU	32°	224°
3		E1	37.3	AEU	66°	208°
3		E5a	33.8	AEU	66°	208°
5		E1	25.3	AE	24°	157°
5		E5a	25.6	AEU	24°	157°
8		E1	33.0	AEU	42°	302°
8		E5a	26.2	AEU	42°	302°
24		E1	45.4	AEU	37°	59°
24		E5a	37.9	AEU	37°	59°
25		E1	43.2	AEU	79°	161°
25		E5a	37.2	AEU	79°	161°
33		E1	25.4	AE	10°	20°
7		B1	15.4	AE	13°	42°
10		B1	26.7	AE	23°	50°
19		B1	41.4	AEU	31°	207°
19		B2a	32.9	AEU	31°	207°
21		B1	29.0	AE	15°	326°
21		B2a	23.4	AE	15°	326°

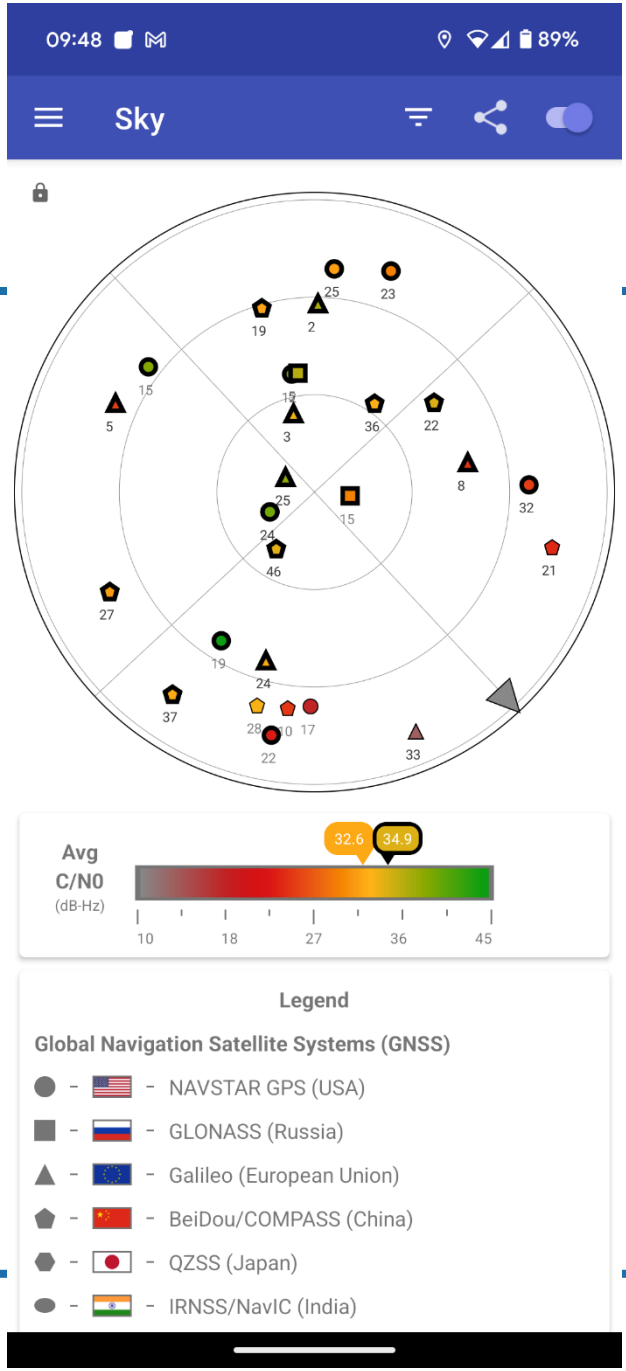
Smartphone GNSS

Sky map of visible GNSS constellations with connected satellites in bold outline.

C/N0 = ratio of signal power to noise power density, is an important parameter for measuring the quality of the received signal.

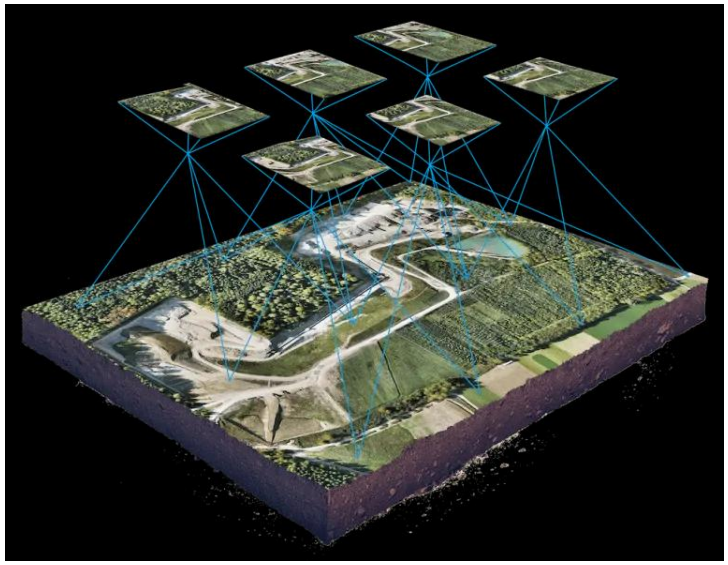
Good Conditions: Values in the 35–45 dB-Hz range are common (satellite elevation, obstructions, antenna, band).

GPStest app downloaded from Google Play App Store.

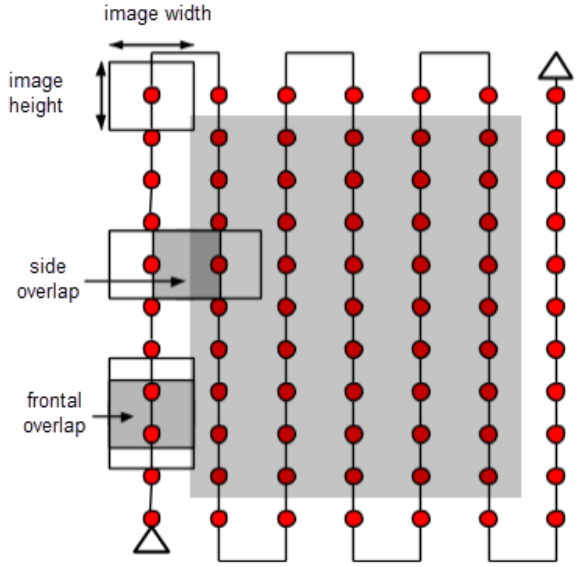


Photogrammetry

- Photogrammetry measures (*x*, *y*, *z*) coordinates of objects using overlapping pairs of aerial photographs (stereo-pair) taken from different points of view.



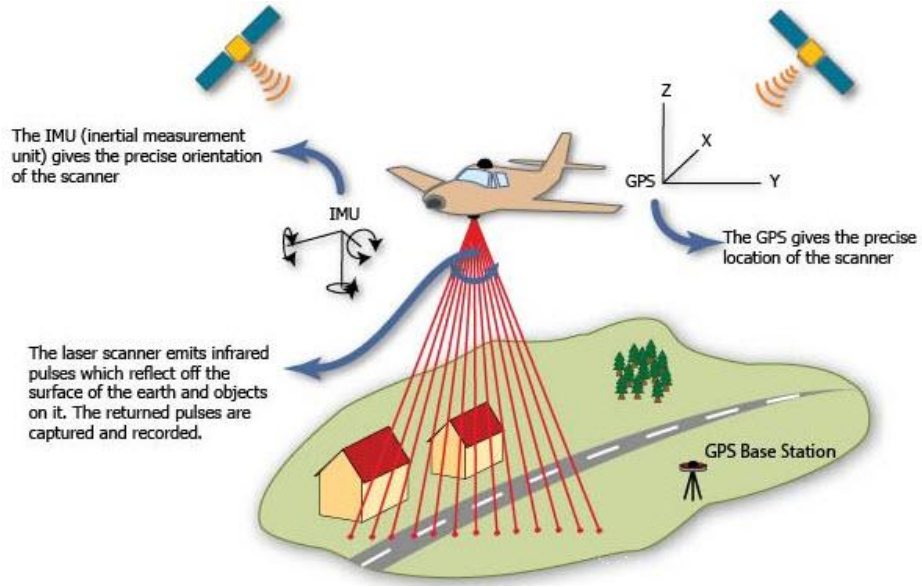
https://wingtra.com/drone-photogrammetry-vs-lidar/



https://www.aerotas.com/overlap-flight-pattern

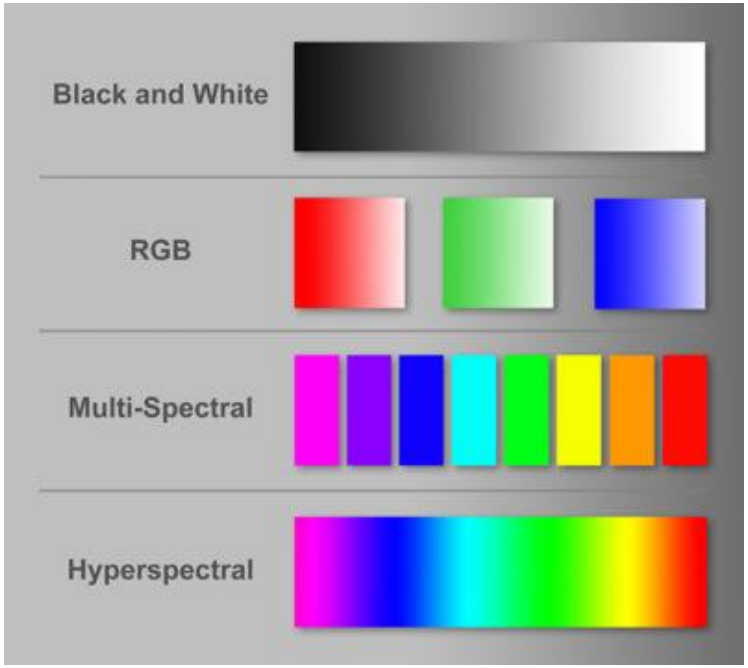
Laser scanning (LiDAR)

- LiDAR produces (*x*, *y*, *z*) coordinates for multiple discrete points, called mass points or point clouds.
- Photogrammetry and LiDAR can achieve a vertical accuracy of 10-15 cm and horizontal accuracy of 50-100 cm.



Raster data capture

- Remotely sensed images, including satellite images and aerial photographs. Four main types visible, infrared, multispectral, and hyperspectral.



Visible (greyscale or colour bands)

Visible and infrared (colour bands or continuous)

Crowdsourcing

- Collect spatial data by soliciting contributions from the community or general public.
- Involve the use of Internet-enabled and location-aware mobile devices to gather location-specific information.
- Disadvantages: possibly skewed data, lack of ownership or confidentiality, loss of expertise.











Spatial data transfer

- The process of converting or translating existing digital spatial data and importing them into a GIS.
- Import existing spatial data from external sources.
- May be in different data formats.
- Read directly or translated into a suitable format.



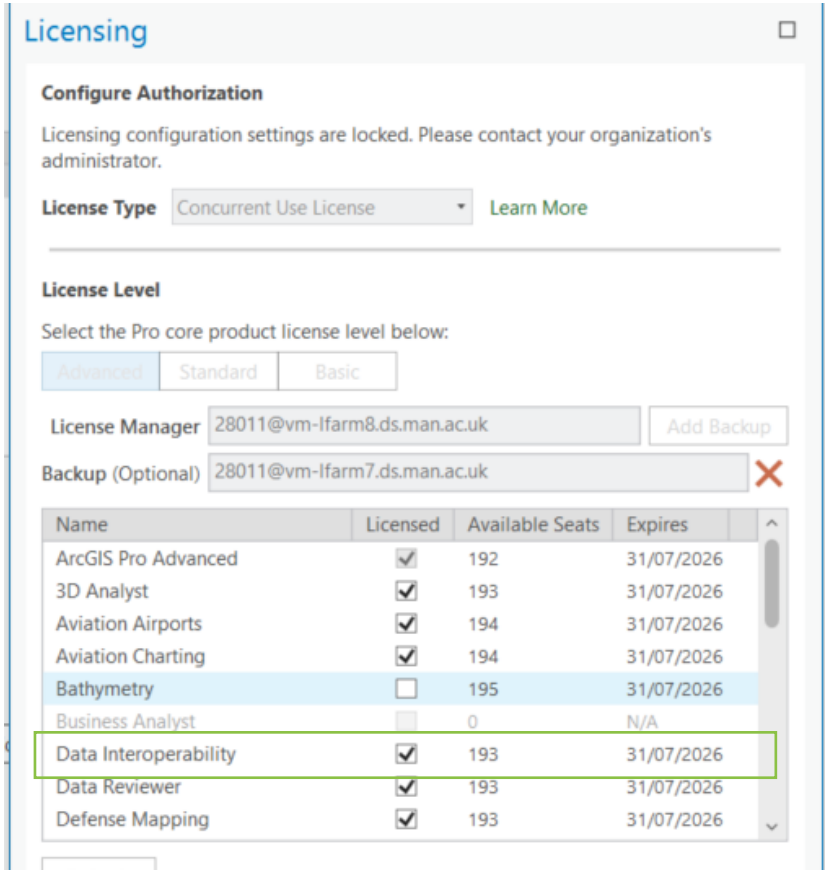
Digital spatial data formats

Vector	Raster
ESRI ArcInfo Coverage	ESRI GRID
ESRI ArcInfo E00 (ArcInfo Interchange Format)	DTED (Digital Terrain Elevation Data)
ESRI ArcView Shapefile	ERDAS IMG (IMAGINE)
MapInfo MIF (Map Interchange Format)	BMP (Windows Bitmap)
USGS DLG (Digital Line Graphs)	TIFF (Tagged Image File Format)
AutoCAD DXF (Data Exchange Format)	GeoTIFF
AutoCAD DWG (Drawing)	GIF (Graphics Interchange Format)
CGM (Computer Graphics Metafile, ISO)	JPEG (Joint Photographic Experts Group)
Microstation DGN (Microstation drawing file format)	PNG (Portable Network Graphics)
SDTS (Spatial Data Transfer Standard , US)	USGS DEM
NTF (National Transfer Format, UK)	ECW (Enhanced Compressed Wavelet)
TIGER (Topologically Integrated Geographic Encoding and Referencing System, US Bureau of Census)	
DIME (Dual Independent Map Encoding, US Bureau of Census)	
VPF (Vector Product Format, National Imagery and Mapping Agency, US)	
KML (Keyhole Markup Language, Google)	
WFS (Web Feature Service, Open Geospatial Consortium)	
NetCDF (Network Common Data Form, Unidata)	
S-57 (International Hydrographic Organization)	

Feature Manipulation Engine

- ArcGIS Pro incorporates FME technology for spatial data transfer.
- Supports >300 formats of spatial and non-spatial data.

e.g. various GIS, rasters, databases, 3D, XML, and point cloud (LiDAR) data.

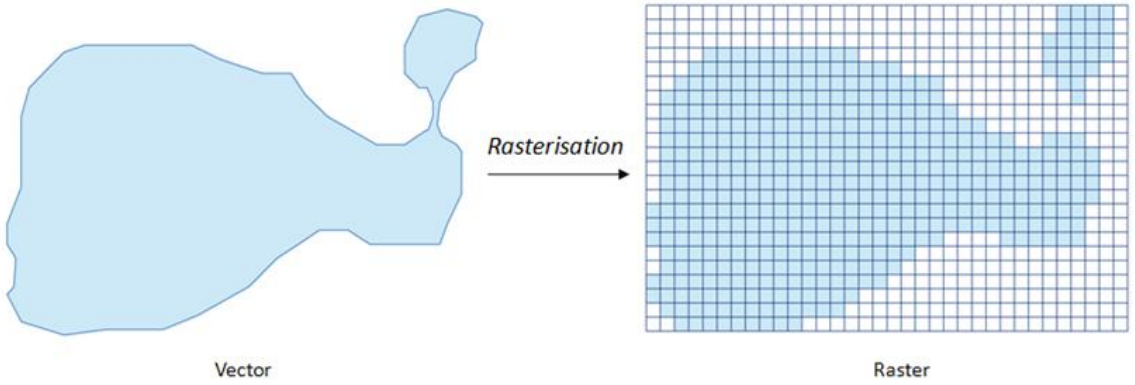


Spatial data transformation and editing

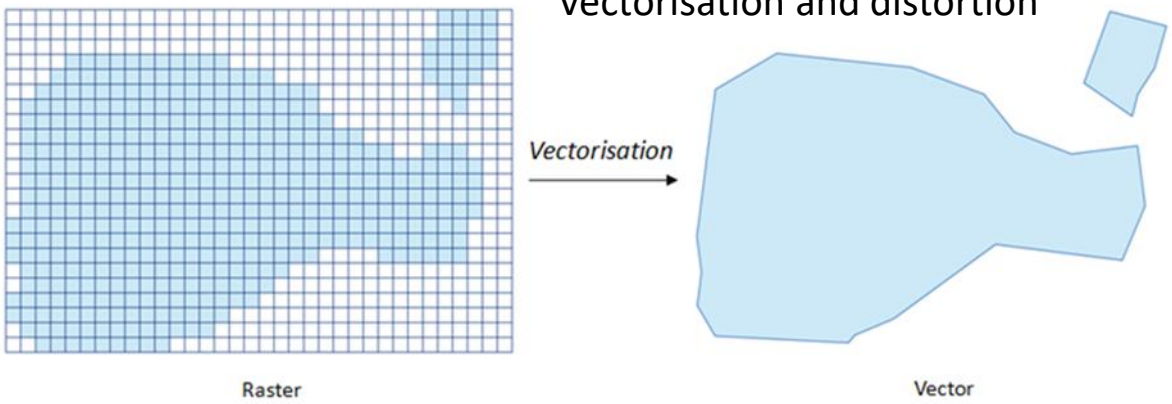
- Data structure conversion (raster-vector conversion).
- Geometric transformation – rubber sheeting.
- Line simplification.
- Edge matching.

Data structure conversion

Rasterisation and loss of information

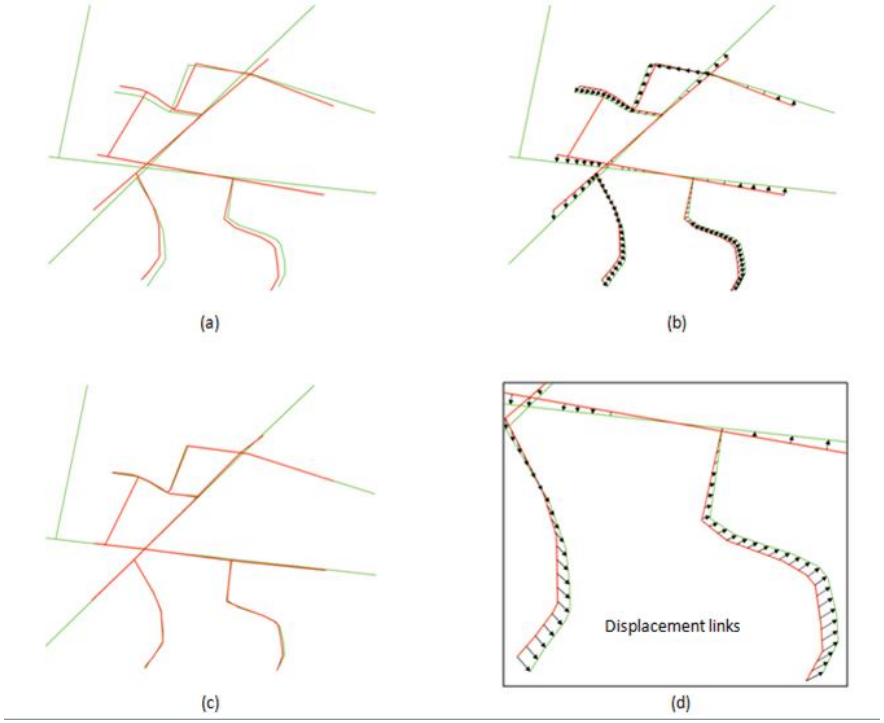


Vectorisation and distortion

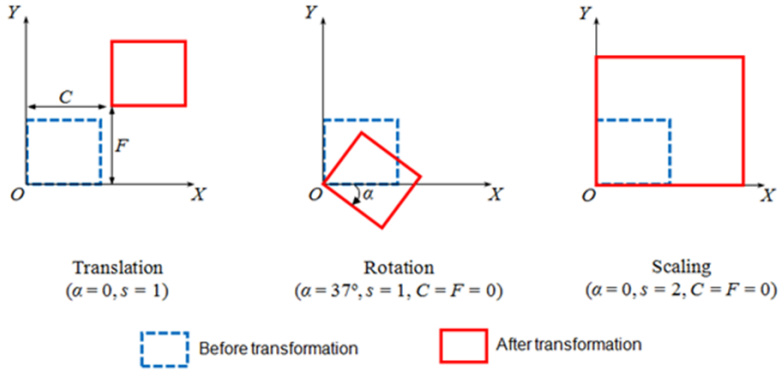


Spatial transformations

Rubber sheeting – correcting geometric distortions

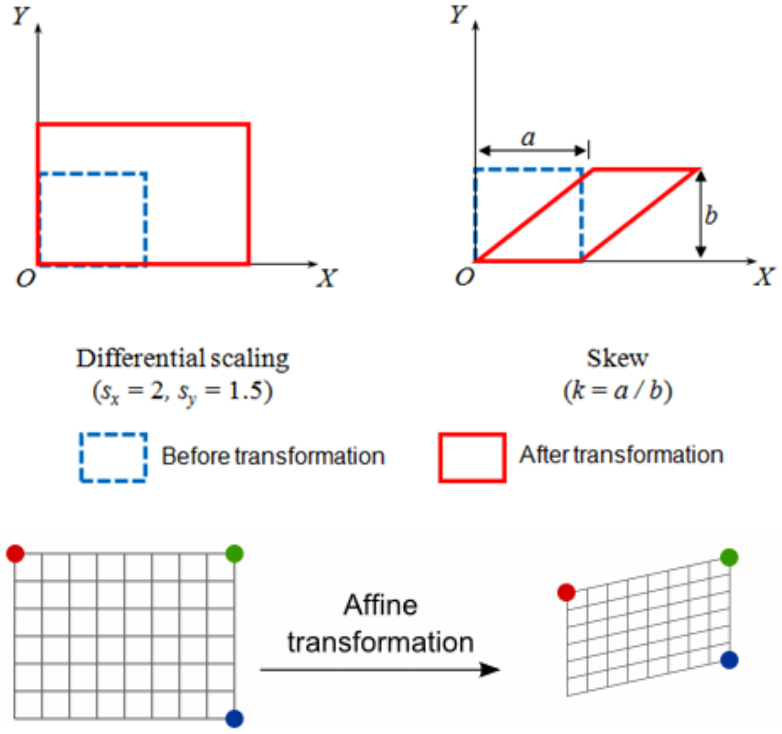


Similarity transformation – translate, rotate or scale – preserves shape (angles do not change)



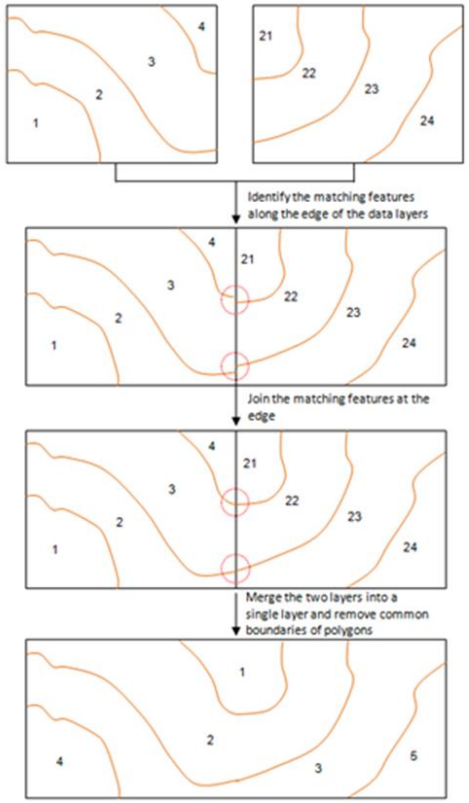
Spatial transformations

Affine transformation scales, skews, rotates and translates – preserve parallelism (length and angles may change).



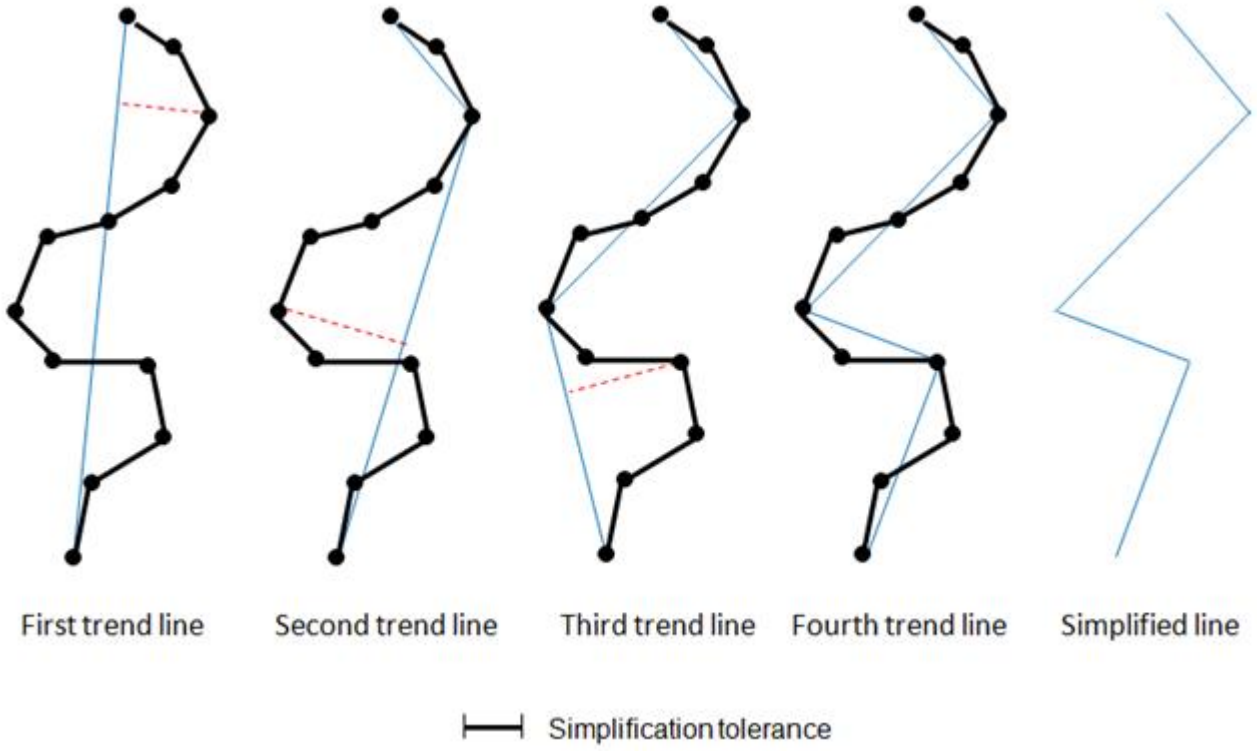
Requires a minimum of 3 control points

Edge matching – match features across data layers.



Line simplification (weeding)

Removes insignificant bends and redundant points or vertices from a line or a polygon while preserving its essential shape.



Spatial data errors

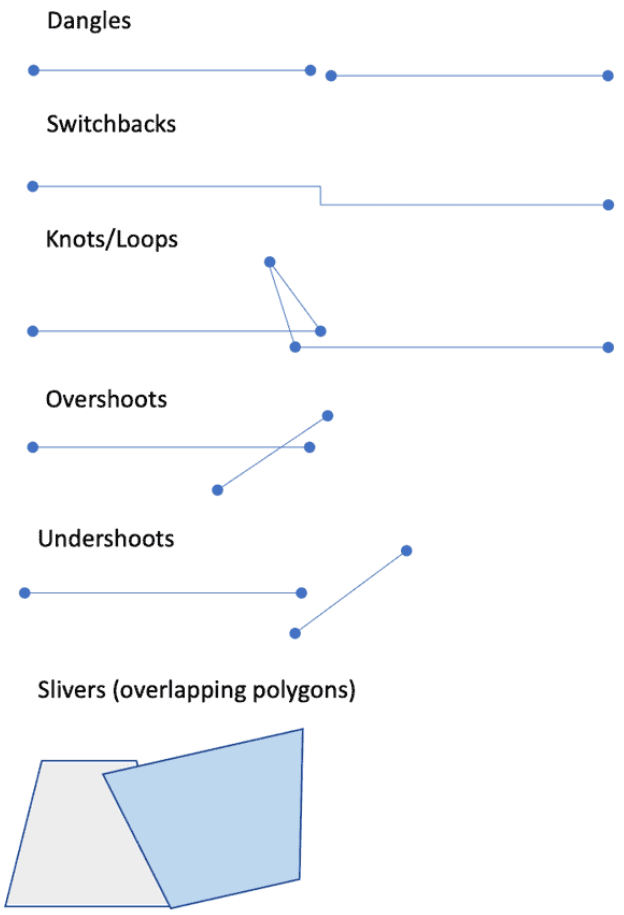
Some common sources of errors in GIS data:

- Data formatting errors
  - scale, coordinate projection, raster conversion etc
- Cartographical generalisation errors
- Location errors
- Topological errors
- Attribute errors

Topological errors

Topology errors arise when spatial features don't connect, intersect or align correctly, according to the spatial rules in a GIS.

Most are common when manually digitising a scanned map (except knots).



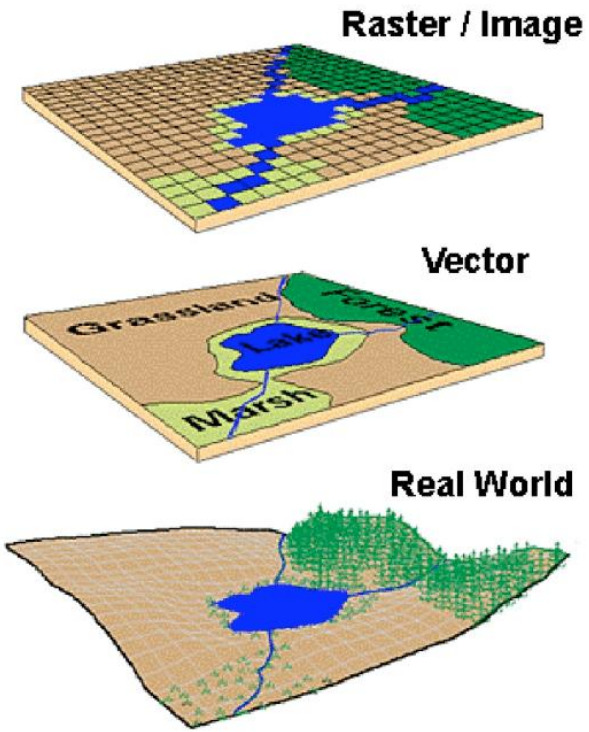
Attribute errors

- Attribute data may contain observational or measurement errors, data entry errors or be simply out of date.
- Editing includes adding, deleting, or updating attributes associated with features and their values.

Spatial data quality

- Spatial data may contain inherent errors.
- Their quality must be compatible with their intended applications.
- *Spatial data quality* - the level of accuracy, precision, completeness, consistency and timeliness of the data that makes them appropriate for a particular use.

Attribute errors and accuracy



Map data

Reference data

How many pixels have been assigned to the incorrect land use and land cover type?

Cross-tabulation or error matrix

Reference data

Map data

	Grassland	Forest	Marsh	Rows Total
Grassland	28	14	15	57
Forest	1	15	5	21
Marsh	1	1	20	22
Columns Total	30	30	40	100

Grassland = (28/30) * 100 = 93%
Forest = (15/30) * 100 = 50%
Marsh = (20/40) * 100 = 50%

Overall Accuracy = (28 + 15+ 20) = 63%

Positional accuracy

- How closely the coordinate descriptions of features represented in the data compare with their true location.
- Measured using the Root Mean Square Error (RMSE): xyz = true location, x'y'z' = encoded position; n = number of data values.

$$RMSE_{(x,y)} = \sqrt{\frac{\sum_{i=1}^{n}(x_i - x'_i)^2 + \sum_{i=1}^{n}(y_i - y'_i)^2}{n}}$$

$$RMSE_z = \sqrt{\frac{\sum_{i=1}^{n}(z_i - z'_i)^2}{n}}$$

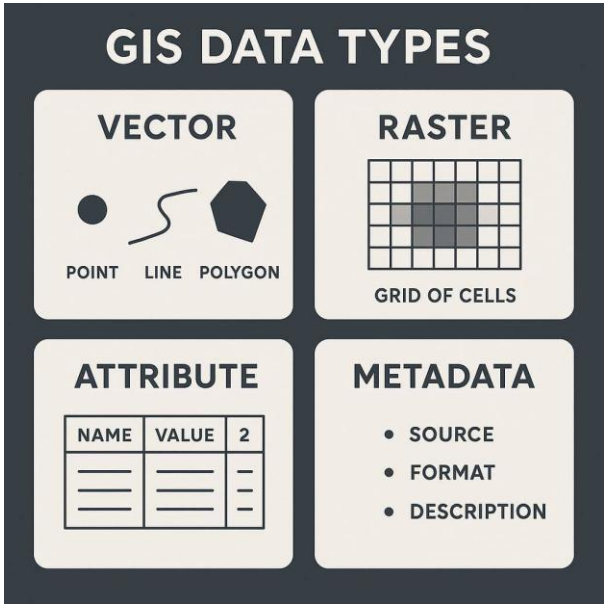
- RMSE has the same units as the x,y,z measurements (e.g. metres, deg-C. etc). An RMSE = 0 means there is no error in the accuracy.

Data quality

- **Precision**: The recorded level of detail.
- **Consistency**: The lack of apparent contradictions between datasets or within a dataset.
- **Completeness**: The extent and range of a dataset with regard to completeness of coverage, classification and verification.

Metadata

- The data about data.
- Describe the content, quality, condition and other characteristics of a dataset.
- Lineage - history of both the source data and the processing steps used to produce the spatial dataset.
- International metadata standards.



Homework and Resources

- Read Chapter 3 of course textbook: Spatial Data Input and Manipulation
- Take the quiz to test your basic understanding (on Canvas)

Further resources

ESRI training: Getting Started with Imagery and Remote Sensing (component 1 only; 45 mins)

https://www.esri.com/training/catalog/634f16c15030f204ae8e30f4/getting-started-with-imagery-and-remote-sensing/